# Multiple models in real-time response

## Comparative COVID-19 modelling

Kath Sherratt
4 March 2026

# Introductions



My background

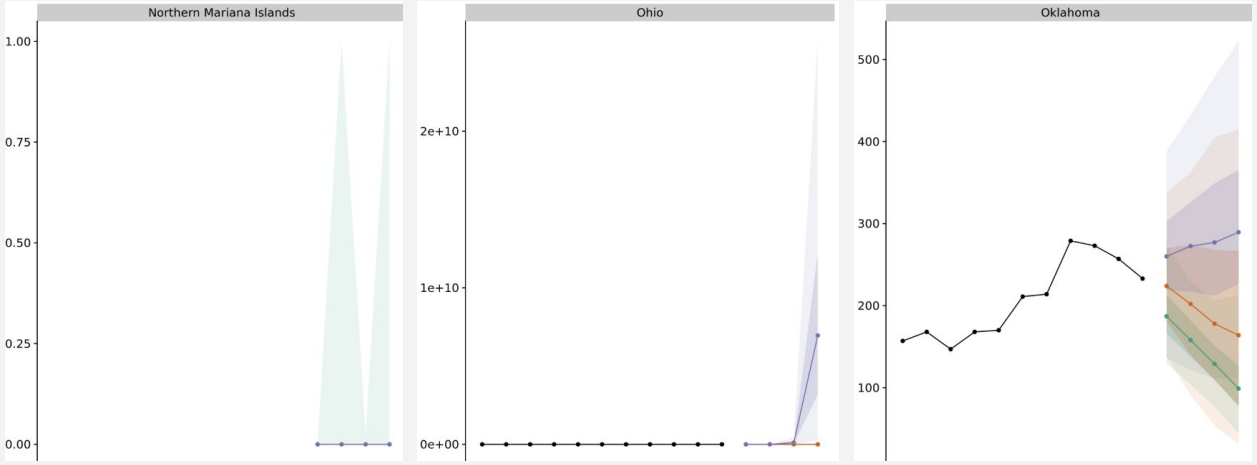
- Work > Epi MSc > Started a PhD > oh dear

- COVID-19: real-time modelling for UK & Europe
  - Rt estimation
  - Forecasting
  - Model comparison & combination

- Comparison = evaluation
  - Why are we so wrong?
  - Can we do any better?

**2.4 Worrying Selectively**

Since all models are wrong the scientist must be alert to what is importantly wrong. It is inappropriate to be concerned about mice when there are tigers abroad. Box 1976




Many ways of being wrong in February 2021

# Modelling in real-time

Many **uncertainties** of an evolving outbreak:

- **Mechanisms**: unobserved, heterogenous transmission
- **Data**: inherently lagged, often sparse and biased
- **Decisions**: timeliness essential, often can't wait for routine surveillance data

Modelling offers **synthesis**:

- expert judgement
- available data
- for a target context

# Modelling in real-time

Many **uncertainties** of an evolving outbreak:

- **Mechanisms**: unobserved, heterogenous transmission
- **Data**: inherently lagged, often sparse and biased
- **Decisions**: timeliness essential, often can't wait for routine surveillance data

Modelling offers **synthesis**:

- expert judgement
- available data
- for a target context

# Modelling in real-time

Modelling offers **synthesis**: *...what is, will, and could happen*

- expert judgement
- available data
- target context

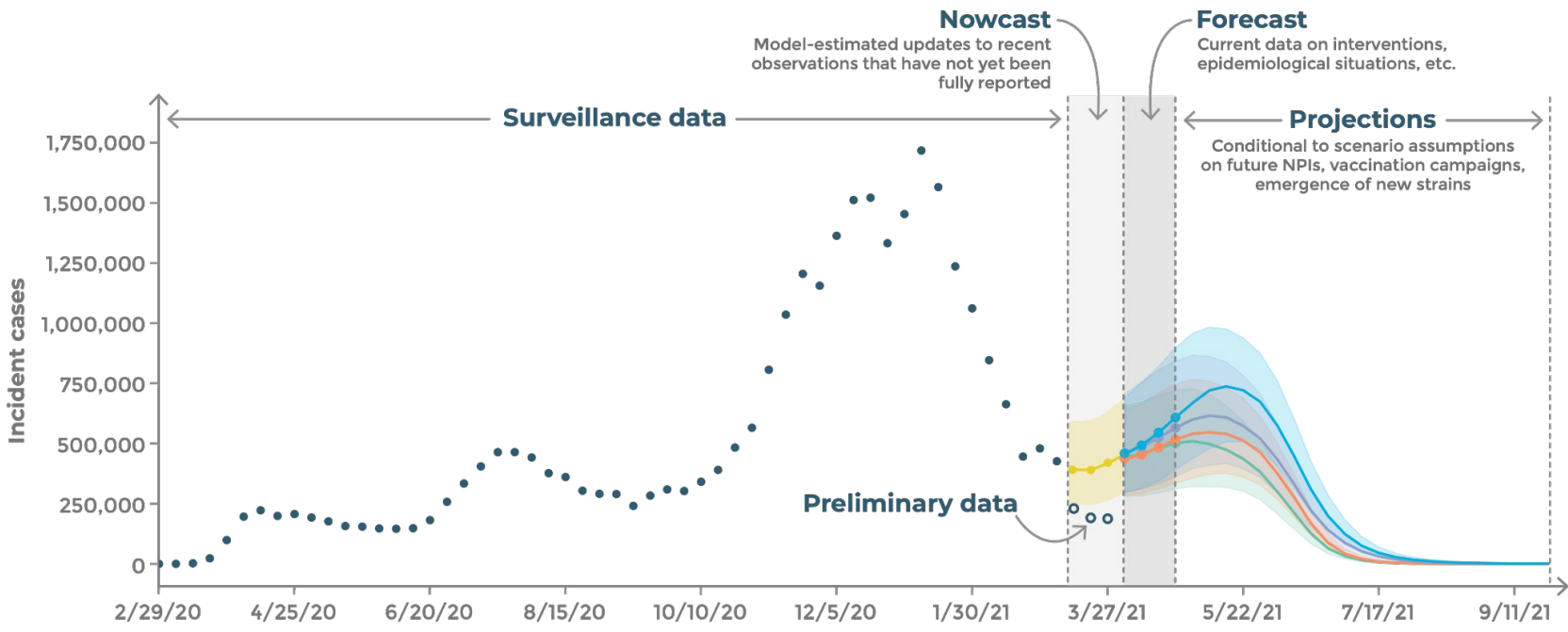



Figure credit: Alex Vespignani and Nicole Samay

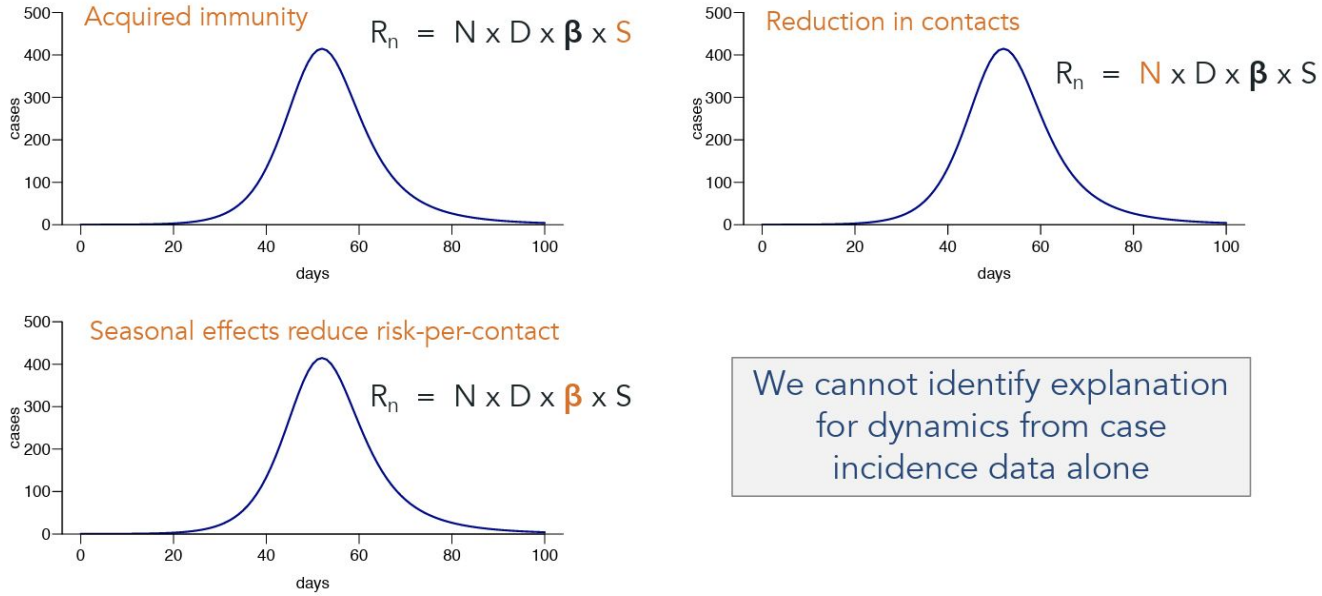
# Modelling in real-time

Modelling offers **synthesis**:

- expert judgement
- available data
- target context

*…This is still uncertain!*

Immunity, seasonality, or control? The problem of identifiability




We cannot identify explanation for dynamics from case incidence data alone

Credit: Adam Kucharski

# Modelling in real-time

Modelling offers **synthesis**:

- expert judgement
- available data
- target context

*…This is still uncertain!*

- Whose judgement?
- Whose data?
- What's needed?

# Modelling in real-time

Modelling offers **synthesis**:

- expert judgement
- available data
- target context

*…This is still uncertain!*

- Whose judgement?
- Whose data?
- What's needed?

*Enter*: **multi-model comparison**

# Comparative modelling in infectious disease epi

LONDON SCHOOL of HYGIENE &TROPICAL MEDICINE

## UK SPI-M

- 2009+: up to a dozen UK modelling groups informing UK government in emergencies
  - Compare, evaluate, form a consensus for policy

## Infectious Disease Modelling Hubs

- Open, accessible platforms for storing, evaluating, and visualising models
  - Anyone can contribute a model
  - Anyone can view model results

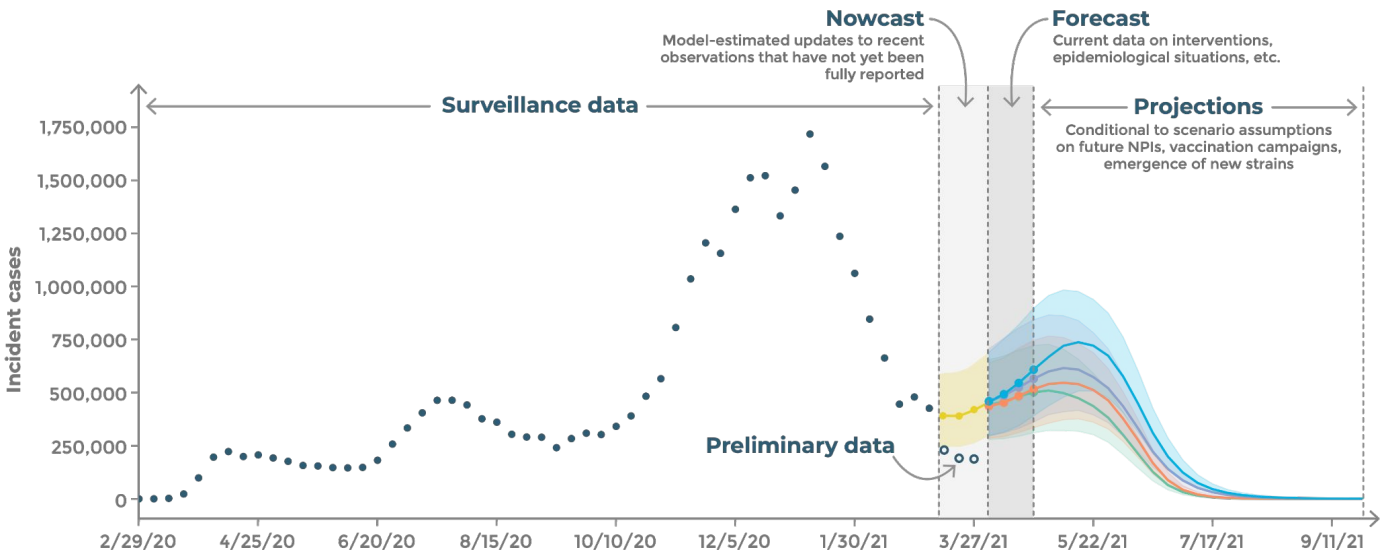



Figure credits: Alex Vespignani and Nicole Samay

# Model comparison in real-time response

Themes

Benefits

*Comparing multiple models yields greater information than any single model*

Detecting dynamics

Decision support

Costs

*Combining multiple models can be misleading and inefficient*

Information loss

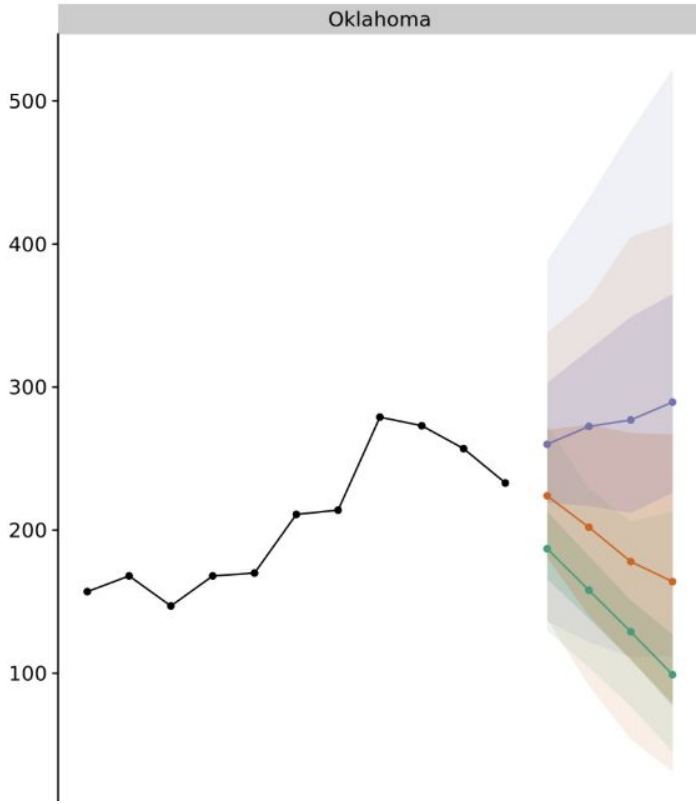
Infrastructure

*When is it **more** useful to compare/combine?*

*When is it **less** useful to compare/combine?*

# Model comparison in real-time outbreak response

What do you pick?




*Many ways of being wrong in February 2021*

LONDON SCHOOL of HYGIENE &TROPICAL MEDICINE

# Model comparison in real-time response

## Themes

### Benefits

*Comparing multiple models yields better information than any single model*

Detecting dynamics

Decision support

### Costs

*Combining multiple models can be misleading and inefficient*

Information loss

Infrastructure

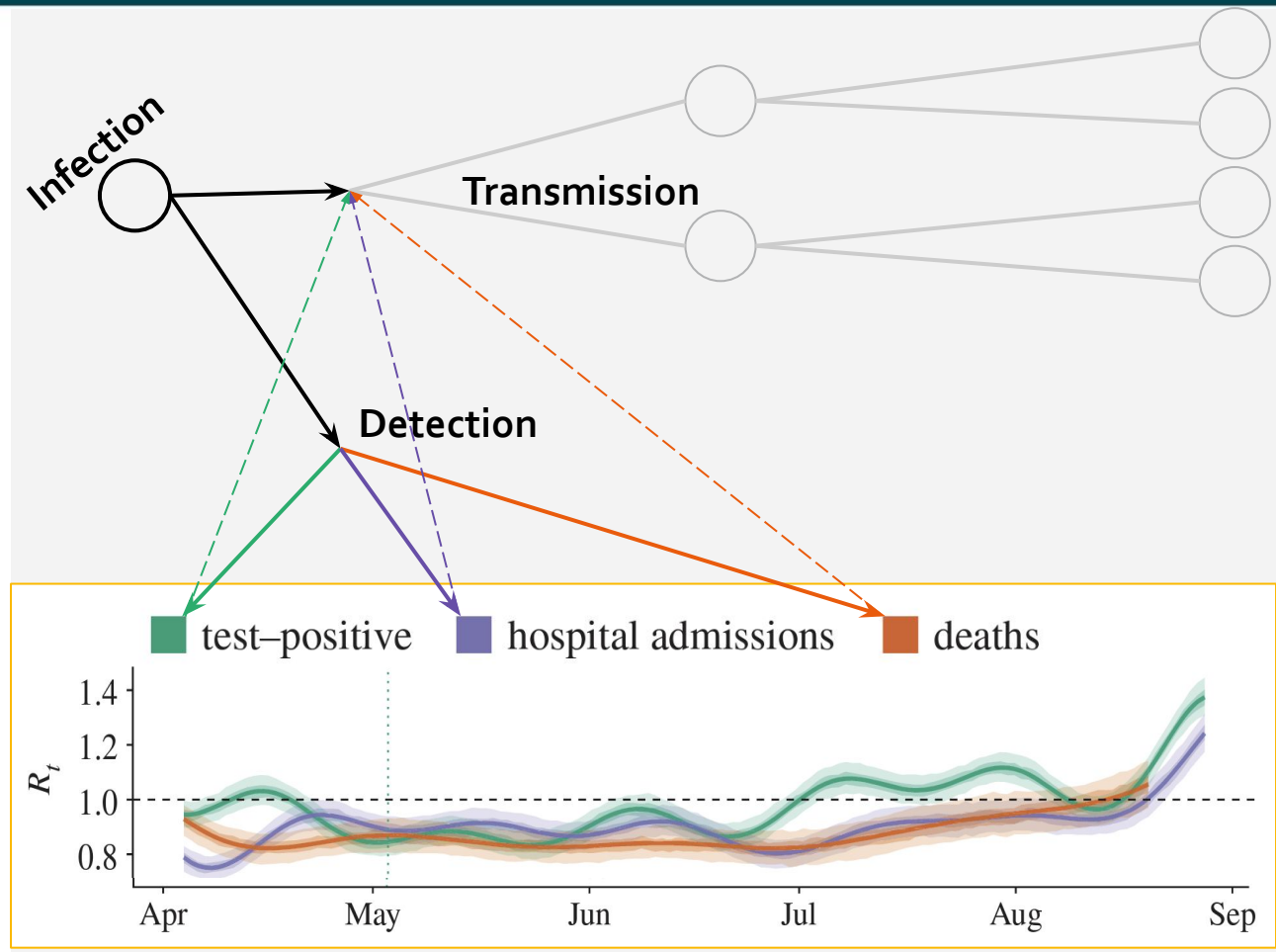
*When is it **more** useful to compare/combine?*

*When is it **less** useful to compare/combine?*

# Benefits of comparison: Detecting dynamics



- Remember the **reproduction number**
  - **Effective Rn** (*net/time-varying*)
  - **Average** onward transmission among a population
- Rarely actually observe infections
  - Use a model to estimate Rn
  - Drawing from data:
    - positive tests (cases)
    - hospital admissions
    - deaths




**COVID-19 effective reproduction number** in England over 2020 by data source

*Sherratt, K.*, Abbott, S.*, and others, 2021. Exploring surveillance data biases when estimating the reproduction number. Phil. Trans. R. Soc., doi.org/10.1098/rstb.2020.0283*

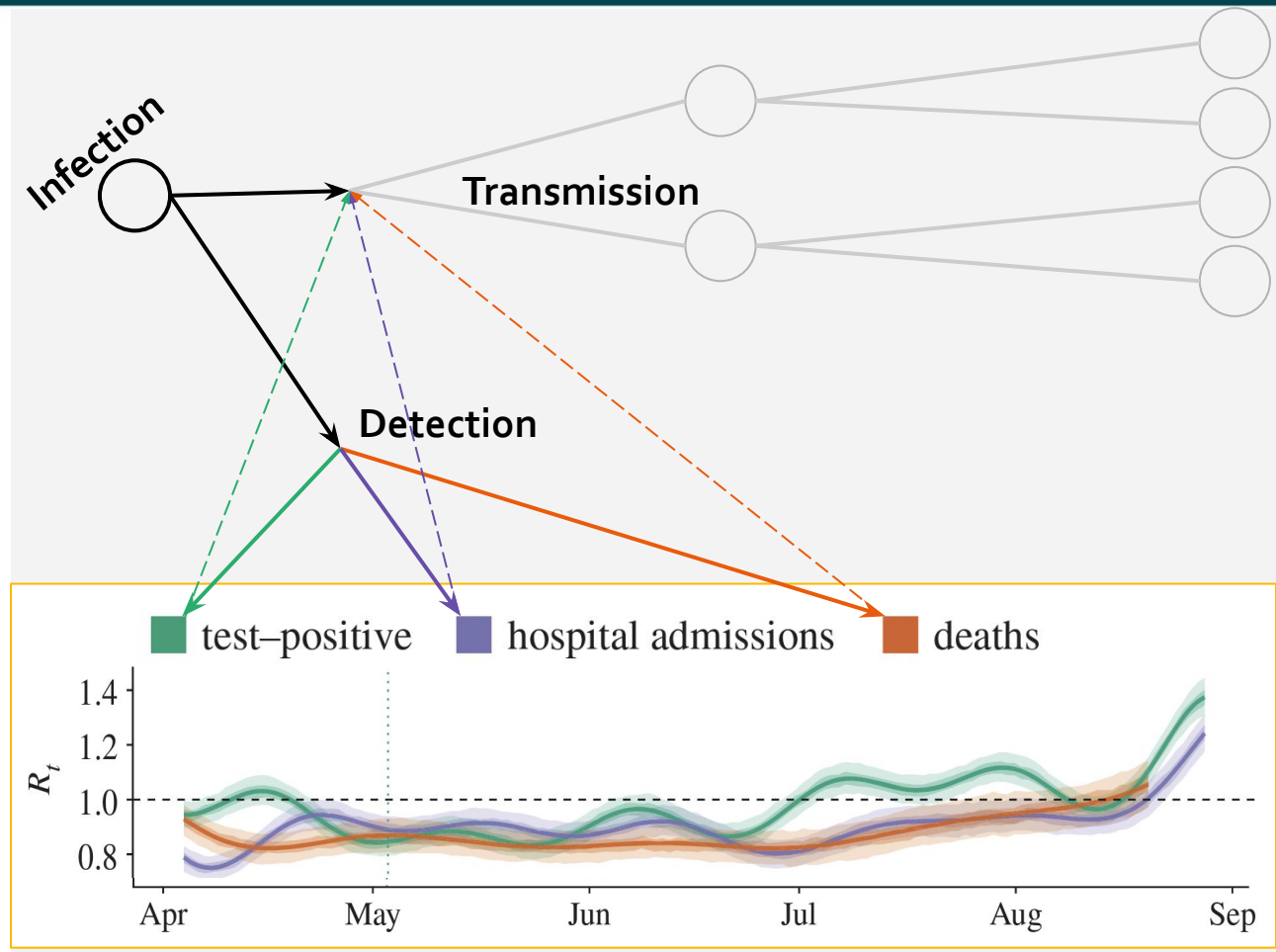
# Benefits of comparison: Detecting dynamics



Each data source is a biased sample of transmission across the population

- **Multiple models:** model Rt separately from each data source
  - Differences may indicate heterogeneous transmission and severity

*…Challenge assumptions*




**COVID-19 effective reproduction number** in England over 2020 by data source

*Sherratt, K.*, Abbott, S.*, and others, 2021. Exploring surveillance data biases when estimating the reproduction number. Phil. Trans. R. Soc., doi.org/10.1098/rstb.2020.0283*

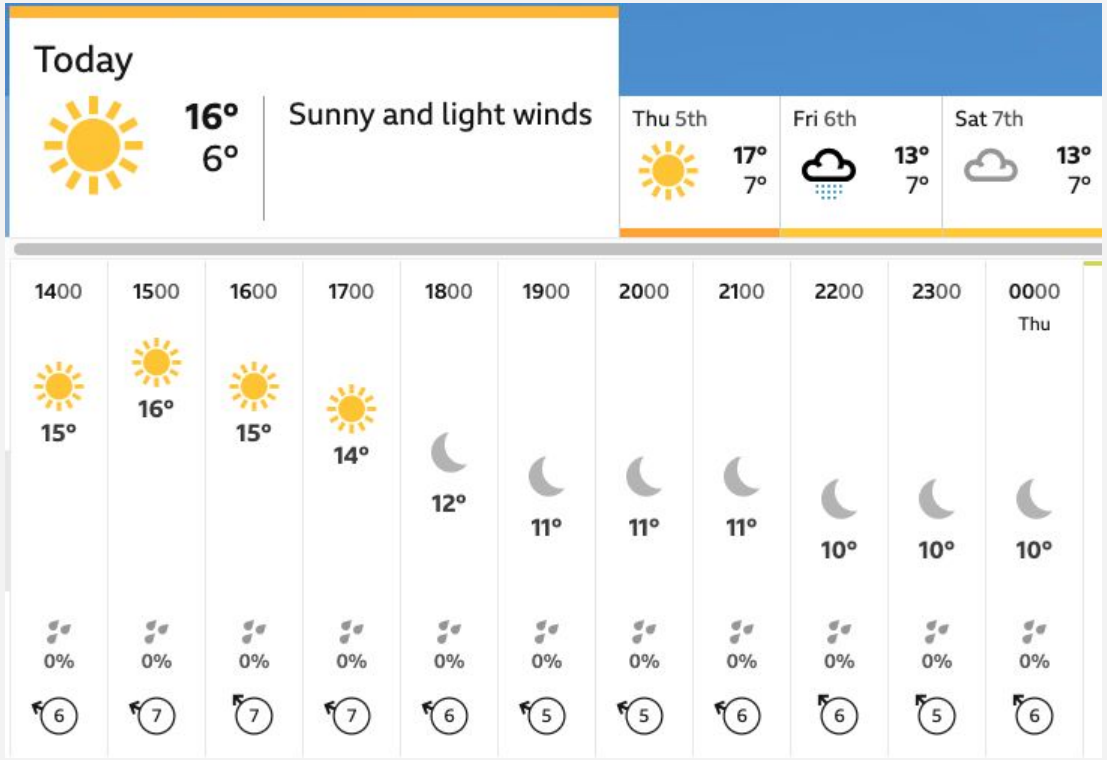
# Benefits of comparison: Predictive accuracy



## Combining predictions

- With many models, we can **combine** the predictions in an **ensemble**
  - Gather multiple prediction from (slightly) different models
  - **Average!**

- Well established finding of more **reliable predictive accuracy**
  - Many other fields work with (quantitative, predictive) models
  - Physics, weather, economics…



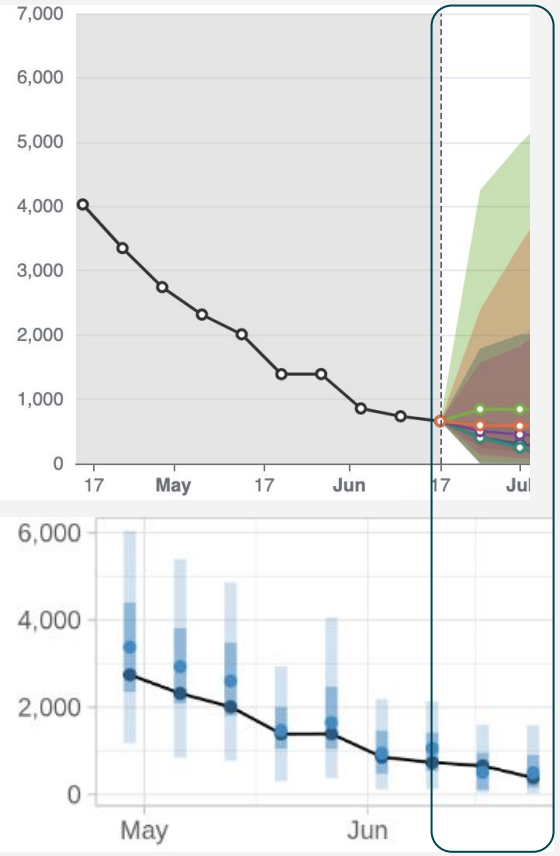

**Weather forecasts** are the combination of many meteorological model results into a single ensemble prediction

# Benefits of comparison: Predictive accuracy



## Combining predictions

- European Forecast Hub
  - Forecasting COVID-19 in 32 countries over 3 years
  - Forecasts from **~50 models**
- Ensemble: median (typical) prediction from all models




**COVID-19 hospitalisations in Germany,** May-July 2023: forecasts from all models made on 17 June 2023 (top); performance of the Hub ensemble (below)

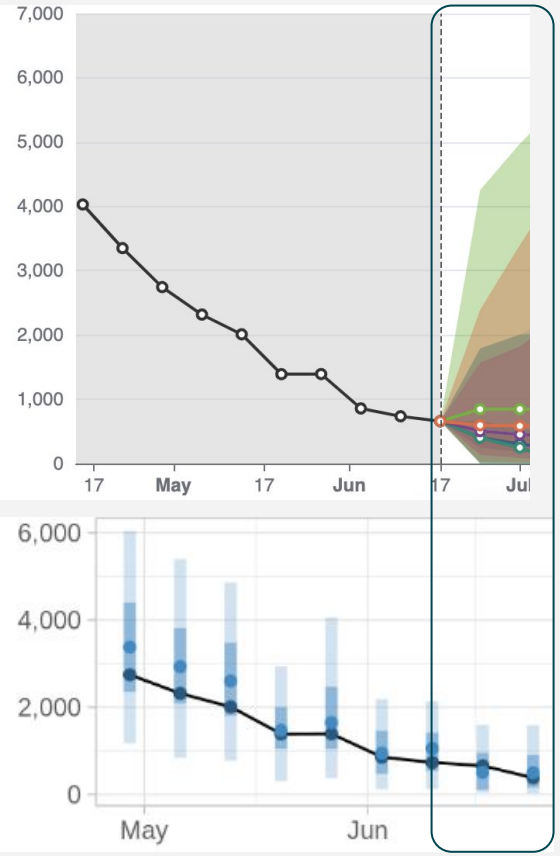
*Sherratt, K., and others, 2023. Predictive performance of multi-model ensemble forecasts of COVID-19 across European nations. eLife, 12:e81916, doi.org/10.7554/eLife.81916*

# Benefits of comparison: Predictive accuracy



## Evaluating predictions

- How long do you expect forecasts to remain accurate?
- How might that vary by the prediction target (i.e. the data stream you are trying to predict)?
- Why?




**COVID-19 hospitalisations in Germany,** May-July 2023: forecasts from all models made on 17 June 2023 (top); performance of the Hub ensemble (below)

*Sherratt, K., and others, 2023. Predictive performance of multi-model ensemble forecasts of COVID-19 across European nations. eLife, 12:e81916, doi.org/10.7554/eLife.81916*

# Benefits of comparison: Predictive accuracy



**Evaluating predictions**

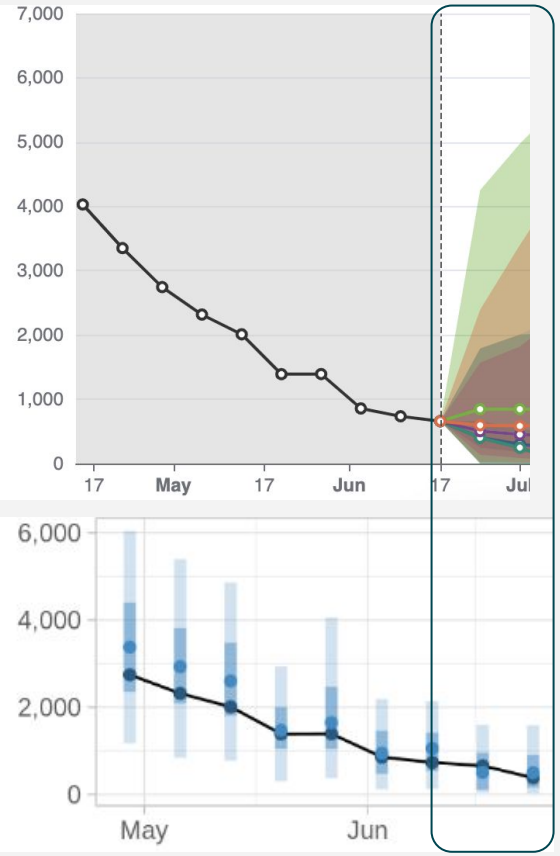
forecasts vs. data

i.e. *model prediction - observed data*

Predictions best when:

- Short term (~2 week) predictions
  - For longer-lagged data (i.e. deaths)
- Ensemble - most reliably accurate

Hypothesis: why **<2-3 weeks**, not longer?




**COVID-19 hospitalisations in Germany**, May-July 2023: forecasts from all models made on 17 June 2023 (top); performance of the Hub ensemble (below)

*Sherratt, K., and others, 2023. Predictive performance of multi-model ensemble forecasts of COVID-19 across European nations. eLife, 12:e81916, doi.org/10.7554/eLife.81916*

# Benefits of comparison: Predictive accuracy



**Evaluating predictions**

forecasts vs. data

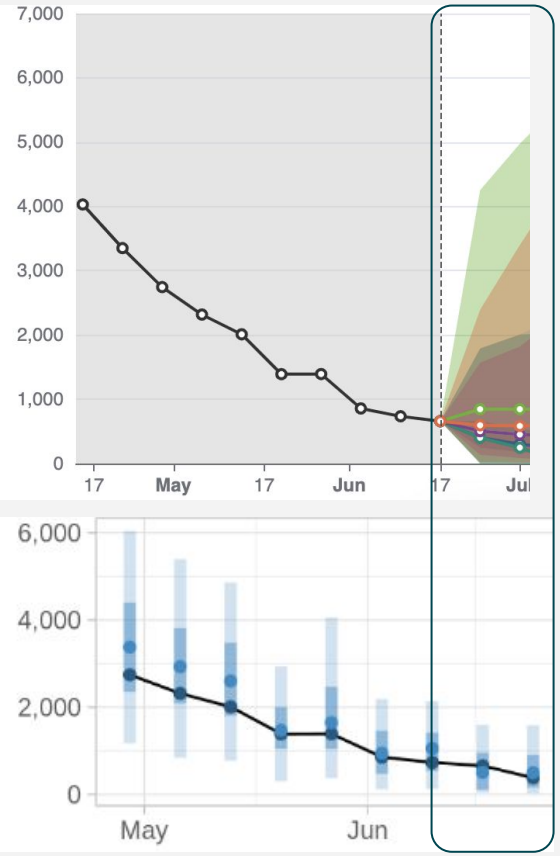
i.e. *model prediction - observed data*

High reliability of ensemble for:

- Short term (~2 week) predictions
- For longer-lagged data (i.e. deaths)

Hypothesis: why **<2-3 weeks**, not longer?

- predicting *transmission* - depends on time-duration of current transmission
  - Serial interval




**COVID-19 hospitalisations in Germany**, May-July 2023: forecasts from all models made on 17 June 2023 (top); performance of the Hub ensemble (below)

*Sherratt, K., and others, 2023. Predictive performance of multi-model ensemble forecasts of COVID-19 across European nations. eLife, 12:e81916, doi.org/10.7554/eLife.81916*

# Benefits of comparison: Predictive accuracy



**Evaluating predictions**

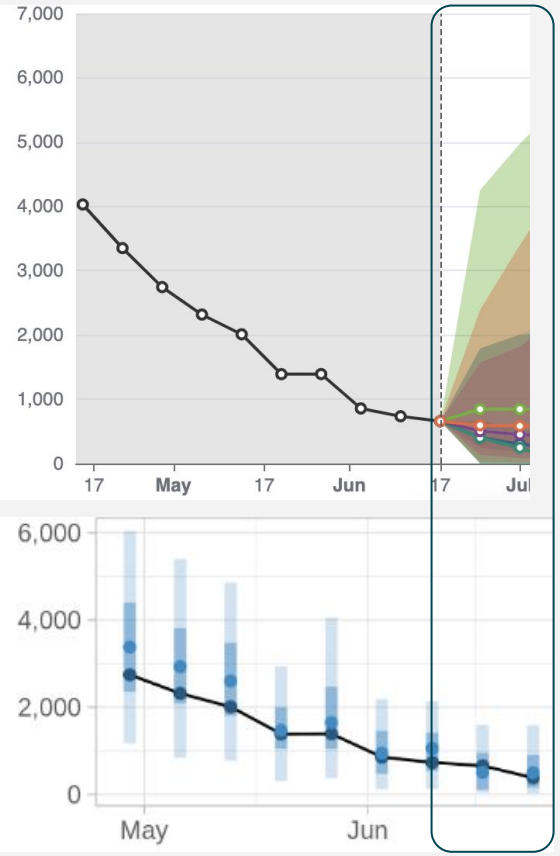
forecasts vs. data

i.e. *model prediction - observed data*

High reliability of ensemble for:

- Short term (~2 week) predictions
- For longer-lagged data (i.e. deaths)

Hypothesis: why **ensemble** more accurate than most individual models?




**COVID-19 hospitalisations in Germany,** May-July 2023: forecasts from all models made on 17 June 2023 (top); performance of the Hub ensemble (below)

*Sherratt, K., and others, 2023. Predictive performance of multi-model ensemble forecasts of COVID-19 across European nations. eLife, 12:e81916, doi.org/10.7554/eLife.81916*

# Benefits of comparison: Predictive accuracy



**Evaluating predictions**

forecasts vs. data

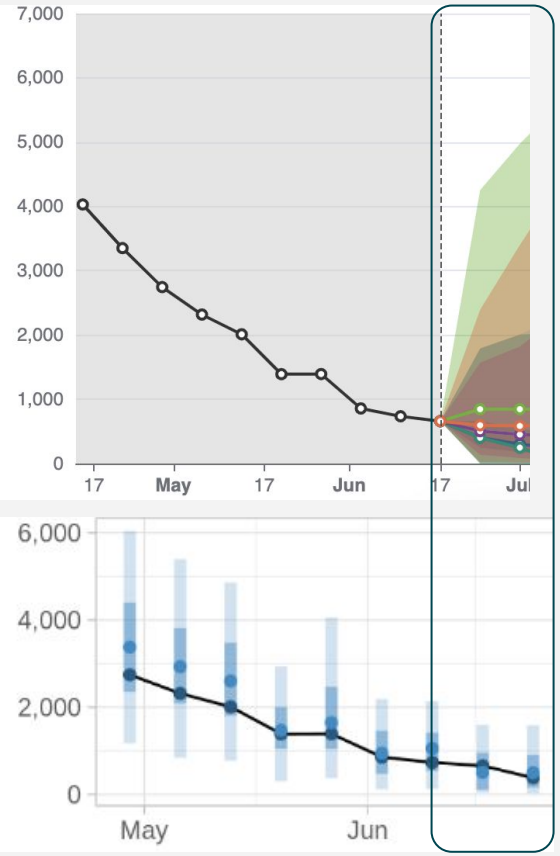
i.e. *model prediction - observed data*

High reliability of ensemble for:

- Short term (~2 week) predictions
- For longer-lagged data (i.e. deaths)

Hypothesis: why **ensemble** more accurate than most individual models?

- Reduced variance
- … "wisdom of crowds"…




**COVID-19 hospitalisations in Germany,** May-July 2023: forecasts from all models made on 17 June 2023 (top); performance of the Hub ensemble (below)

*Sherratt, K., and others, 2023. Predictive performance of multi-model ensemble forecasts of COVID-19 across European nations. eLife, 12:e81916, doi.org/10.7554/eLife.81916*

LONDON SCHOOL of HYGIENE &TROPICAL MEDICINE

# Model comparison in real-time response

## Themes

### Benefits

*Comparing multiple models yields greater information than any single model*

Detecting dynamics

Decision support

### Costs

*Combining multiple models can be misleading and inefficient*

Information loss

Infrastructure

*When is it **more** useful to compare/combine?*

*When is it **less** useful to compare/combine?*

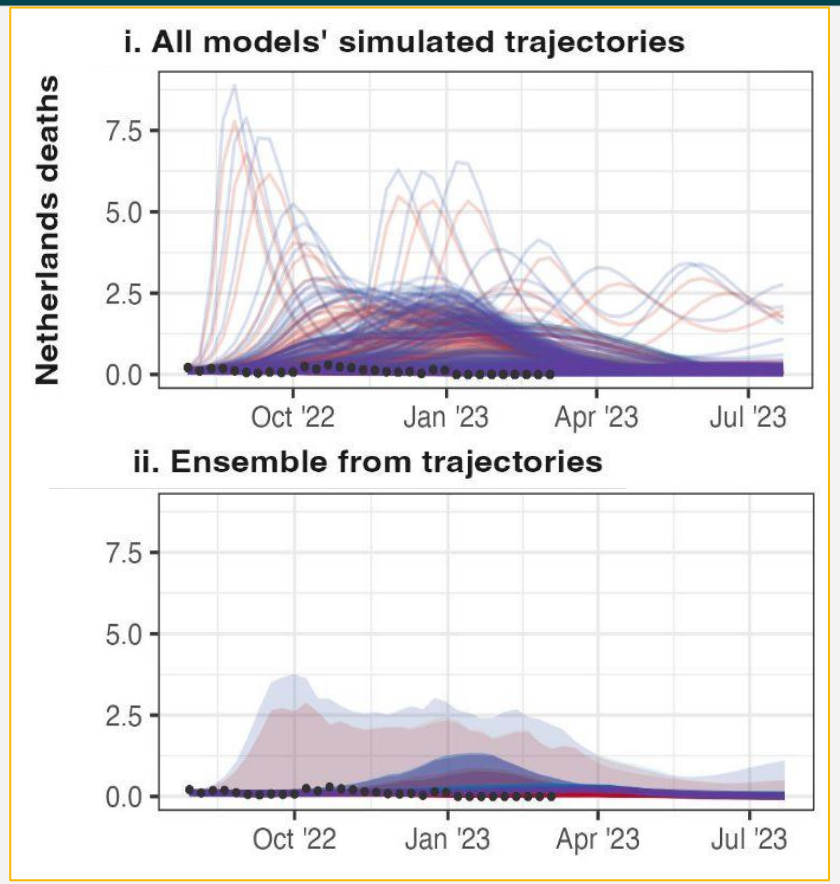
# Costs of comparison: Information loss



**Aggregation dilutes detail**

- Combining temporal trajectories smoothes out variation between models
  - **Ensemble model doesn't actually represent any single projection**
  - Combined estimate may be **uninformative of any real process**

When might this matter?




**Different epidemic shapes seen in (i) c.300 model trajectories or (ii) a median ensemble.** Projections for Netherlands incident deaths per 100,000, across 4 scenarios, from 3 models

*Sherratt, K., and others, 2023. Characterising information loss due to aggregating epidemic model outputs. medRxiv, under review at Epidemics. doi.org/10.1101/2023.07.05.23292245*

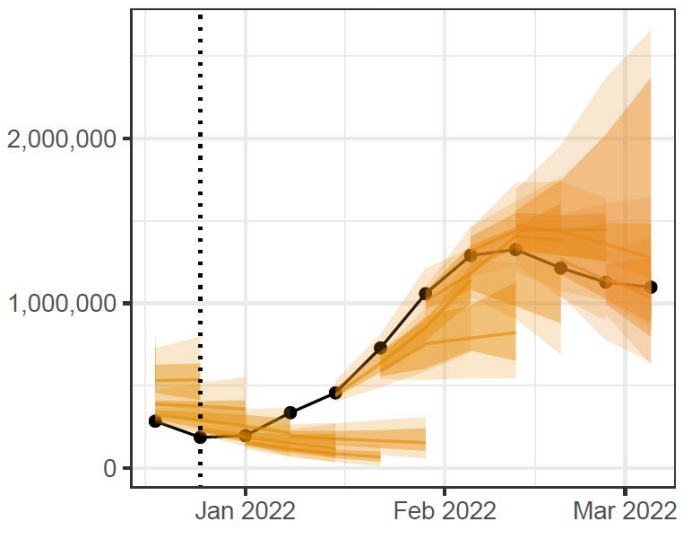
# Costs of comparison: Unknown unknowns



**Aggregation dilutes detail**

- Combining temporal trajectories smoothes out variation between models
  - **Ensemble model doesn't actually represent any single projection**
  - Combined estimate may be **uninformative of any real process**

When might this matter?




**Cases in Germany: difficult to predict Omicron variant introduction,** observed cases (black), and European Forecast Hub ensemble (orange)

# Costs of diversity: Infrastructure



**Institutional structures**

- Appropriate comparison requires some
  - Standardisation - to be clear what's varying between models
  - Facilitation and translation of "consensus"
- Responsive real-time work is fundamentally different from standard long-term research
  - Relies on **trust** and communication

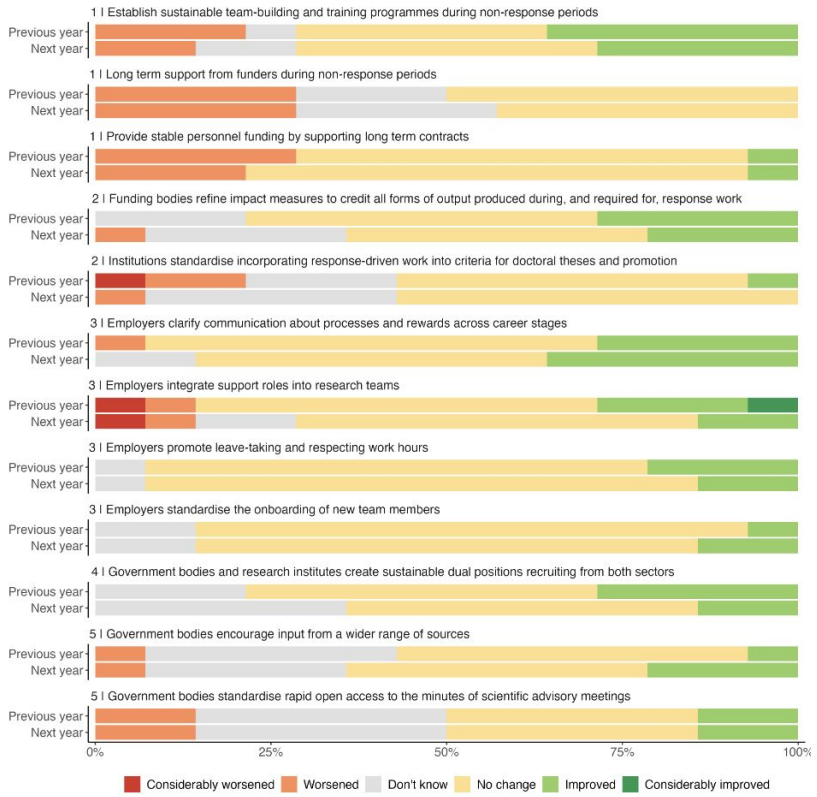



**Figure 1.** Views of recommended actions for outbreak response modelling as of April 2024 among 14 survey responders in the UK modelling community. Respondents were asked whether each situation had improved or worsened over the past year, and what their expectations were for the next year, across five priority areas with 12 recommended actions for improvement.

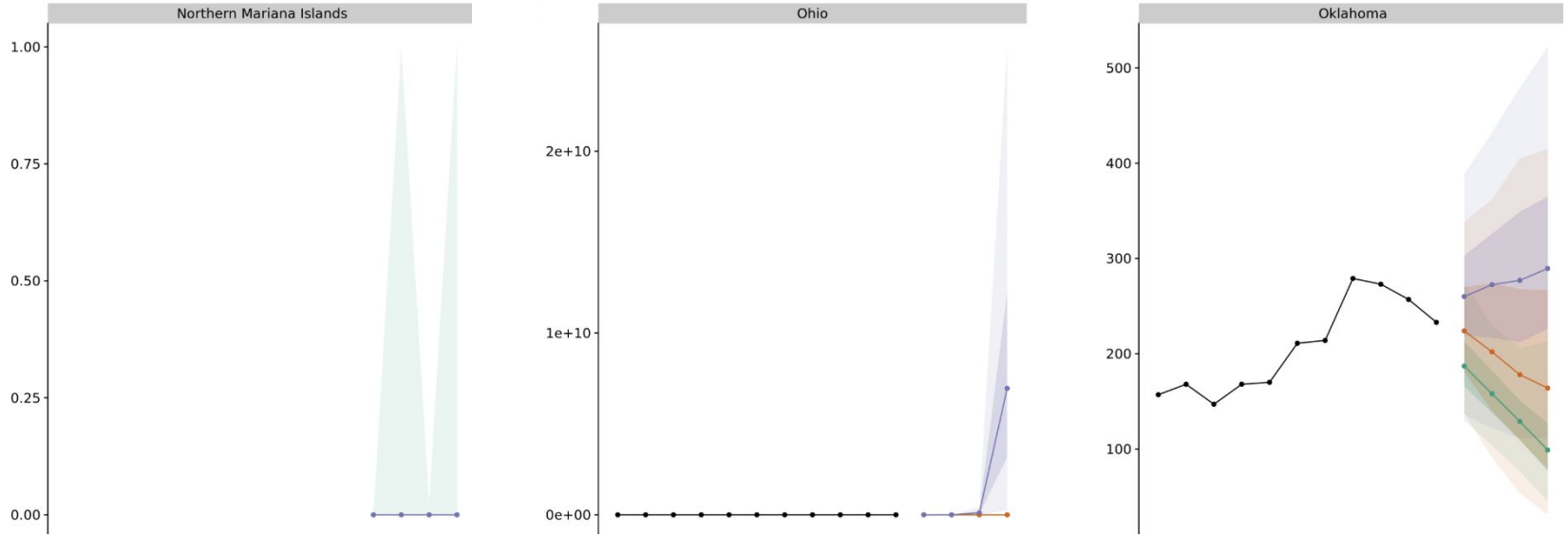
*Sherratt, K.*, Carnegie, A.*, and others, 2023. Improving modelling for epidemic responses. bioRxiv, under review at Wellcome Open Research. doi.org/10.1101/2023.06.12.544667*

# Evaluating modelling in real-time outbreak response

## 2.4 Worrying Selectively

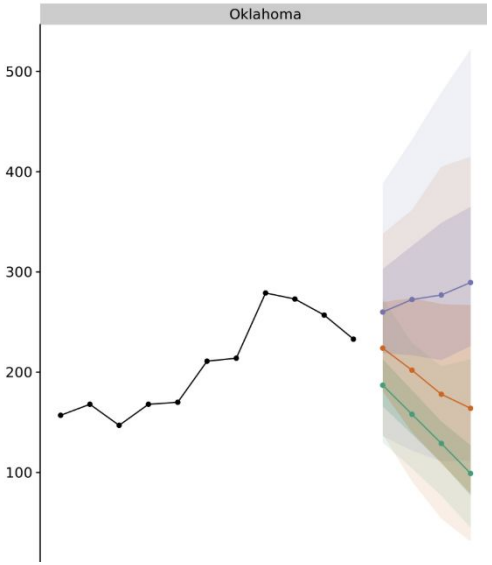
Since all models are wrong the scientist must be alert to what is importantly wrong. It is inappropriate to be concerned about mice when there are tigers abroad.

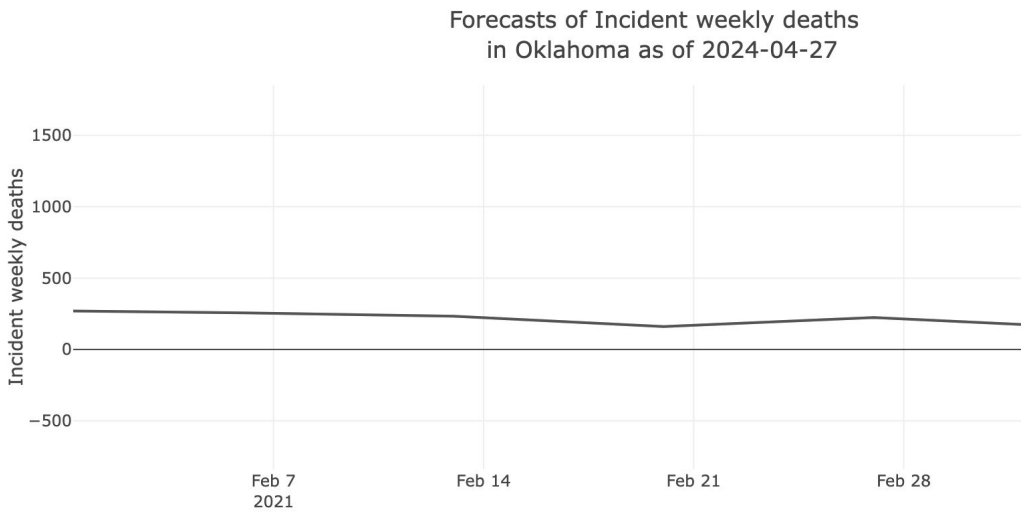
Box 1976




Many ways of being wrong in February 2021

# Evaluating modelling in real-time outbreak response




Many ways of being wrong in February 2021

LONDON SCHOOL of HYGIENE &TROPICAL MEDICINE

# Acknowledgements



Contact: Katharine Sherratt

- katharine.sherratt@lshtm.ac.uk
- More on some of this work:
- https://kathsherratt.github.io/

Very many thanks to collaborators and colleagues

- Sebastian Funk, Sam Abbott, and the epiforecasts team
- CMMID COVID-19 Working Group
- ECDC
- Over 50 teams participating and providing modelling results in real-time

# Further reading: wider perspectives on modelling


**Van Kerkhove MD, Ferguson NM. Epidemic and intervention modelling--a scientific rationale for policy decisions? Lessons from the 2009 influenza pandemic. Bull World Health Organ. 2012 Apr 1;90(4):306–10. doi:10.2471/BLT.11.097949 PubMed PMID: 22511828; PubMed Central PMCID: PMC3324871.**

**Leach M, Scoones I. The social and political lives of zoonotic disease models: Narratives, science and policy. Social Science & Medicine. 2013 Jul 1;88:10–7. doi:10.1016/j.socscimed.2013.03.017**

Metcalf CJE, Edmunds WJ, Lessler J. Six challenges in modelling for public health policy. Epidemics. 2015 Mar 1;Challenges in Modelling Infectious DIsease Dynamics10:93–6. doi:10.1016/j.epidem.2014.08.008

Boden LA, McKendrick IJ. Model-Based Policymaking: A Framework to Promote Ethical “Good Practice” in Mathematical Modeling for Public Health Policymaking. Frontiers in Public Health. 2017;5:68. doi:10.3389/fpubh.2017.00068

Redd SC, Frieden TR. CDC’s Evolving Approach to Emergency Response. Health Security. 2017 Feb;15(1):41–52. doi:10.1089/hs.2017.0006

Nixon K, Jindal S, Parker F, Reich NG, Ghobadi K, Lee EC, et al. An evaluation of prospective COVID-19 modelling studies in the USA: from data to science translation. The Lancet Digital Health. 2022 Oct 1;4(10):e738–47. doi:10.1016/S2589-7500(22)00148-0 PubMed PMID: 36150782.

**Teerawattananon Y, Kc S, Chi YL, Dabak S, Kazibwe J, Clapham H, et al. Recalibrating the notion of modelling for policymaking during pandemics. Epidemics. 2022 Mar;38:100552. doi:10.1016/j.epidem.2022.100552 PubMed PMID: 35259693; PubMed Central PMCID: PMC8889889.**

Oliwa J, Guleid FH, Owek CJ, Maluni J, Jepkosgei J, Nzinga J, et al. Framework to guide the use of mathematical modelling in evidence-based policy decision-making. BMJ Open. 2025 Apr 5;15(4):e093645. doi:10.1136/bmjopen-2024-093645 PubMed PMID: 40187784; PubMed Central PMCID: PMC11973756.


**[Some top picks in bold]**

# Further reading: references for this talk


Box G. Robustness in the Strategy of Scientific Model Building. In: Robustness in Statistics. Academic Press; 1979 [cited 2026 Mar 4]. p. 201–36. Available from: https://www.sciencedirect.com/science/chapter/edited-volume/abs/pii/B9780124381506500182 doi:10.1016/B978-0-12-438150-6.50018-2

Sherratt K, Abbott S, Meakin SR, Hellewell J, Munday JD, Bosse N, et al. Exploring surveillance data biases when estimating the reproduction number: with insights into subpopulation transmission of COVID-19 in England. Philosophical Transactions of the Royal Society B: Biological Sciences. 2021 Jul 19;376(1829):20200283. doi:10.1098/rstb.2020.0283

Medley GF. A consensus of evidence: The role of SPI-M-O in the UK COVID-19 response. Adv Biol Regul. 2022 Dec;86:100918. doi:10.1016/j.jbior.2022.100918 PubMed PMID: 36210298; PubMed Central PMCID: PMC9525209.

Reich NG, Lessler J, Funk S, Viboud C, Vespignani A, Tibshirani RJ, et al. Collaborative Hubs: Making the Most of Predictive Epidemic Modeling. Am J Public Health. 2022 Jun;112(6):839–42. doi:10.2105/AJPH.2022.306831

Sherratt K, Gruson H, Grah R, Johnson H, Niehus R, Prasse B, et al. Predictive performance of multi-model ensemble forecasts of COVID-19 across European nations. Wesolowski A, Ferguson NM, Shaman JL, Pei S, editors. eLife. 2023 Apr 21;12:e81916. doi:10.7554/eLife.81916

Sherratt K, Carnegie A, Kucharski A, Cori A, Pearson C, Jarvis C, et al. Improving modelling for epidemic responses: reflections from members of the UK infectious disease modelling community on their experiences during the COVID-19 pandemic. Wellcome Open Research. 2024;9(12). doi:10.12688/wellcomeopenres.19601.1